

NeMo Curator Parquet Boundary Adapter

Source-pinned local engineering evidence for the real ImageWriterStage Parquet writer boundary at NVIDIA-NeMo/Curator@15cc645cbf9e9314fed9e11fc89f6535ea9a8820.

Review request: validate this narrow boundary and route the repository to the appropriate NeMo Curator maintainer for technical review. This is an engineering handoff, not a recruiting request.

Integration boundary

A dedicated source-pinned adapter; no upstream runtime fork.

ImageWriterStage

Pinned at 15cc645cbf9e

Parquet sidecar

Five required writer columns plus
two optional documented scores

Boundary adapter

nemo.image_writer.parquet.source-
first.v1

VULCA handoff

CuratorSignal -> ReviewPacket ->
HumanGate -> EvidenceLedger

Verified evidence

- 87 public tests
- 2 source-first rows
- 2 documented-enriched rows
- 4 JSONL compatibility rows
- Adapter contract: `nemo.image_writer.parquet.source-first.v1`

Why this boundary

The adapter reads one metadata sidecar, validates the pinned writer schema, preserves sidecar and row provenance, and reuses VULCA's existing CuratorSignal, ReviewPacket, HumanGate, and EvidenceLedger path. It does not fork or replace NeMo Curator.

Repository: <https://github.com/vulca-org/vulca-nemo-curator-adapter>

Evidence and limits

Contract and mapping

Required fields are validated; optional values are never invented.

Required

image_id, member_name, metadata,
original_path, tar_file

Optional

aesthetic_score, nsfw_score

Mapping

image_id -> asset_id;
tar_file#member_name -> uri;
source_url|url -> source_url;
caption|text -> caption

Provenance

sidecar filename, row number, raw
metadata, parse status

Reproduce

- Install with `python3.11 -m pip install -e '[nemo-parquet,dev]'`.
- Build both Parquet fixtures from their readable JSON sources.
- Run the three allowlisted test files and verify 87 passing tests.
- Triage the source-first fixture and inspect the manifest, trace, safety checks, and review packets.

Reproduction evidence

Every public count maps to a shipped source or command.

Inspect

2 + 2 Parquet source rows; 4 JSONL
rows

Build

Regenerate both Parquet fixtures
from readable JSON

Test

87 tests in three allowlisted
files

Triage

Run the source-first fixture and
inspect generated evidence

Claim boundary

- Synthetic public-safe fixtures only.
- No full NeMo Curator, CUDA, DALI, Ray, Xenna, GPU, multi-shard, or tar-member execution claim.
- No generic NeMo or Cosmos compatibility claim.
- No NVIDIA endorsement or upstream integration is claimed.

Maintainer review questions

- Is the pinned five-column writer boundary represented accurately?
- Is optional documented-score handling an appropriate forward-compatible boundary?
- Which NeMo Curator owner or contribution surface should evaluate this adapter next?